	Liver	Testis	Bone marrow	Kidney	Islet	Eye	Cortex	Lung	q	S_g	profile correlation	π	dN/dS
Recurrent, above the prior (42)													
Receptors (class)	+7.1*	+7.3*	+9.5*	+15.0*	+13.6*	+7.2*	+13.1*	+14.1*	<.001	8↑0↓	7↑0↓	−2.2	+0.9
Epithelial–mesenchymal transition	+7.8*	+7.6*	+4.0*	+12.1*	+16.7*	+10.0*	+8.1*	+11.6*	<.001	8↑0↓	6↑0↓	+0.4	0.0
Cytoskeleton & ECM (category)	+5.4*	+10.1*	+5.5*	+8.5*	+10.1*	+7.8*	+3.3*	+9.5*	<.001	8↑0↓	6↑1↓	+2.8	−2.9
KRAS signaling, up	+2.9*	+5.1*	+5.2*	+6.5*	+9.8*	+6.1*	+4.5*	+8.9*	<.001	6↑0↓	5↑0↓	−2.7	+1.5
Inflammatory response	+3.7*	+3.4*	+6.4*	+6.3*	+9.9*	+4.0*	+6.1*	+9.1*	<.001	7↑0↓	6↑0↓	−4.2	+3.4
Apical junction	+4.0*	+2.6*	+3.6*	+7.1*	+9.2*	+6.4*	+5.0*	+7.6*	<.001	8↑0↓	6↑0↓	+2.0	−3.2
TNF-α signaling via NF-κB	+1.3	+3.7*	+5.0*	+6.7*	+11.1*	+2.8*	+5.0*	+9.6*	<.001	6↑0↓	6↑0↓	−1.4	+1.5
GPCRs (class)	+3.7*	+4.5*	+1.9	+7.6*	+8.0*	+3.9*	+7.5*	+5.9*	<.001	7↑0↓	6↑0↓	−0.1	−0.5
Coagulation	+4.7*	+4.5*	+2.5*	+6.7*	+8.2*	+4.2*	+4.7*	+6.5*	<.001	7↑0↓	6↑0↓	−1.8	+1.4
IL-2/STAT5 signaling	+2.7*	+3.2*	+5.1*	+5.5*	+7.4*	+5.6*	+4.1*	+7.3*	<.001	7↑0↓	5↑0↓	−1.5	+1.0
Myogenesis	+4.1*	+4.3*	+2.3*	+6.2*	+7.4*	+5.5*	+4.8*	+6.3*	<.001	8↑0↓	6↑0↓	+0.5	−1.2
UV response, down	+3.0*	+4.1*	+1.3	+5.2*	+8.5*	+5.3*	+5.9*	+5.3*	<.001	6↑0↓	7↑0↓	+2.4	−1.9
Complement	+4.5*	+1.4	+4.8*	+5.5*	+7.5*	+3.8*	+4.5*	+6.3*	<.001	7↑0↓	3↑0↓	−0.9	+0.6
Hypoxia	+3.0*	+2.1	+3.8*	+6.1*	+8.0*	+3.9*	+3.5*	+6.3*	<.001	6↑0↓	4↑0↓	+0.7	−1.3
Estrogen response, early	−0.5	+1.8	+2.3*	+7.7*	+7.5*	+5.1*	+4.2*	+7.1*	<.001	5↑0↓	6↑0↓	0.0	−0.3
Allograft rejection	+2.6*	+1.3	+6.7*	+4.5*	+6.3*	+2.9*	+2.2*	+7.0*	<.001	5↑0↓	7↑0↓	−3.0	+2.6
Angiogenesis		+3.1*	+3.5*	+5.1*	+6.1*	+2.6*	+4.4*	+6.3*	<.001	7↑0↓	4↑0↓	−0.1	+0.1
Signaling pathways (category)	+4.8*	+4.4*	+1.1	+2.7*	+5.9*	+4.4*	+2.4*	+7.4*	<.001	5↑0↓	6↑0↓	+6.4	−7.7
Ion channels (class)	−0.7	+7.0*	0.0	+5.9*	+4.9*	+2.6*	+9.4*	+3.2*	<.001	4↑0↓	3↑1↓	+2.9	−5.0
Estrogen response, late	+0.2	+0.8	+3.4*	+7.0*	+6.3*	+4.2*	+3.4*	+6.9*	<.001	6↑0↓	3↑0↓	−0.2	−0.6
Immunity & inflammation (category)	+2.3*	+3.0*	+3.9*	+4.3*	+3.6*	+3.6*	+2.8*	+7.2*	<.001	4↑0↓	6↑0↓	−8.3	+7.0
Apoptosis	+1.5	+2.7*	+4.5*	+4.0*	+6.3*	+2.1	+2.8*	+4.9*	<.001	6↑0↓	4↑0↓	−0.9	+1.2
Protein kinases (class)	+2.9*	+5.0*	+3.2*	+2.1	+5.2*	+0.3	+1.2	+8.3*	<.001	3↑0↓	6↑0↓	+7.0	−7.4
IL-6/JAK/STAT3 signaling	+2.6*	+1.2	+3.9*	+4.0*	+4.4*	+2.7*	+1.5	+5.0*	<.001	6↑0↓	6↑0↓	−4.8	+3.7
Homeodomain TFs (class)	+1.8	+2.5*	−0.3	+4.5*	+5.4*	+2.0	+5.8*	+3.0*	<.001	5↑0↓	6↑0↓	+3.1	−2.7
Interferon-γ response	+1.7	+2.1	+4.3*	+3.4*	+5.3*	+1.8	+0.4	+5.6*	<.001	4↑0↓	4↑0↓	−6.6	+5.3
Androgen response	+2.2	+1.8	+2.2*	+4.6*	+3.2*	+1.1	+4.6*	+4.0*	<.001	4↑0↓	0↑0↓	+0.4	−0.6
Glycolysis	+2.6*	+1.3	+2.1	+4.3*	+3.3*	+3.4*	+2.1	+2.3*	<.001	5↑0↓	1↑0↓	+1.6	−1.2
Xenobiotic metabolism	−1.1	−0.2	+3.2*	+6.9*	+4.2*	+2.5*	+2.2*	+2.8*	<.001	5↑0↓	1↑0↓	−1.1	+0.2
Hedgehog signaling		+2.5*	+1.5	+2.8*	+4.0*	+2.1	+3.0*	+3.1*	<.001	4↑0↓	1↑0↓	+1.7	−2.0
Apical surface		+1.3	+2.7*	+3.6*	+3.3*		+3.6*	+3.0*	<.001	5↑0↓	2↑0↓	−1.2	+0.5
Mitotic spindle	+2.9*	+2.5*	+4.4*	−0.7	+4.3*	+1.6	+1.1	+4.0*	<.001	2↑0↓	3↑0↓	+1.1	−2.0
TGF-β signaling	+2.4*	+1.4	+0.7	+1.2	+5.7*	+1.5	+2.3*	+4.3*	<.001	3↑0↓	5↑0↓	+2.0	−3.3
Nuclear receptors (class)	+1.3	+4.7*	+0.6	+5.9*	+0.8	+1.9	+0.6	+3.6*	0.002	2↑0↓	1↑0↓	+3.0	−2.8
SLC transporters (class)	−0.3	+5.3*	+2.0	+8.8*	+1.0	+1.3	+1.1	−0.2	0.002	2↑0↓	2↑0↓	+0.5	+0.5
KRAS signaling, down	+1.3	+2.2	+1.6	+5.6*	+0.1	+1.4	+3.1*	+2.4*	<.001	4↑0↓	3↑0↓	+0.2	−1.2
p53 pathway	+1.9	−0.6	+2.9*	+1.9	+4.2*	+0.3	+2.4*	+4.4*	<.001	5↑0↓	1↑0↓	+0.2	+0.1
Cholesterol homeostasis	−0.5	+0.9	+1.1	+2.5*	+2.7*	+3.2*	+2.7*	+1.9	<.001	4↑0↓	0↑2↓	−1.8	+1.3
Transcription factors (class)	+5.9*	+3.9*	−2.2*	−0.6	+1.0	+0.4	+1.1	+4.4*	0.002	3↑0↓	5↑0↓	+1.4	−1.4
Notch signaling		−0.1	+1.1	+1.5	+3.0*	+2.7*	+0.9	+2.4*	0.002	2↑0↓	0↑0↓	+2.7	−4.5
mTORC1 signaling	−1.7	−1.4	+5.6*	+2.3*	+3.8*	−0.1	+0.6	0.0	0.002	2↑0↓	1↑1↓	+3.4	−2.7
UV response, up	−0.3	−1.4	+2.7*	+2.6*	+1.6	+0.5	+0.3	+2.1*	0.002	2↑0↓	0↑0↓	+1.5	−1.8
Recurrent, below the prior (15)													
G2/M checkpoint	+1.0	+0.1	+7.2*	−6.1*	+0.5	−1.2	−4.1*	−3.2*	0.002	1↑1↓	1↑1↓	+3.0	−2.4
C2H2 zinc-finger TFs (class)	+3.2*	+0.1	−4.2*	−3.2*	−4.4*	−1.3	−2.9*	+0.6	<.001	0↑4↓	0↑1↓	−1.9	+1.0
Unfolded protein response	−0.7	−5.3*	+2.7*	−3.4*	+1.9	−2.2	−3.3*	−3.0*	<.001	2↑2↓	2↑3↓	+3.6	−2.5
MYC targets V2	−0.9	−4.6*	+2.5*	−2.9*	−0.5	−1.7	−1.8	−3.8*	0.002	1↑1↓	1↑2↓	−1.7	+1.7
E2F targets	+0.1	−3.5*	+9.0*	−6.4*	−2.7*	−4.0*	−3.4*	−4.1*	<.001	1↑4↓	1↑3↓	+1.9	−1.7
Mitochondria & OXPHOS (category)	−3.2*	−3.1*	−4.7*	+1.7	−3.1*	+1.2	−5.0*	−7.6*	<.001	1↑6↓	0↑3↓	−5.0	+3.1
DNA repair	0.0	−6.7*	−0.1	−2.8*	−3.6*	−2.2	−4.5*	−4.8*	<.001	0↑5↓	0↑4↓	+2.2	−3.2
MYC targets V1	−0.6	−7.2*	+4.3*	−4.4*	−2.3*	−4.5*	−5.0*	−5.0*	<.001	1↑3↓	2↑4↓	+7.0	−6.6
Protein homeostasis (category)	−1.9	−4.7*	−1.8	−3.4*	−1.6	−5.5*	−5.8*	−2.7*	<.001	0↑4↓	0↑4↓	+4.1	−3.8
Oxidative phosphorylation	−2.6*	−9.3*	−3.1*	+1.6	−3.0*	+0.7	−5.3*	−9.4*	<.001	1↑5↓	0↑4↓	−1.7	+2.7
Enzymes (class)	−6.2*	−2.0	+0.2	−4.7*	−7.3*	−1.4	−5.3*	−4.8*	<.001	0↑5↓	0↑6↓	+5.0	−2.8
Translation & ribosome (category)	+0.4	−15.8*	0.0	+2.0	−3.9*	−5.6*	−4.2*	−11.2*	<.001	2↑4↓	1↑3↓	−0.5	+1.8
Chromatin regulators (class)	−3.7*	−1.1	−3.4*	−12.0*	−5.5*	−4.6*	−6.5*	−3.4*	<.001	0↑7↓	1↑6↓	+5.3	−4.9
Chromatin & DNA processing (category)	−1.7	−7.2*	−0.8	−9.0*	−6.5*	−5.7*	−8.7*	−8.8*	<.001	0↑5↓	0↑5↓	+4.3	−4.1
Mitochondrial proteins (class)	−5.0*	−14.2*	−6.0*	−0.4	−10.6*	−2.9*	−8.8*	−21.0*	<.001	0↑7↓	0↑6↓	−8.3	+9.0
Called in one tissue only (3)													
Heme metabolism	−1.5	−1.4	+8.2*	+0.3	+0.5	+1.1	+0.4	+0.1	0.351	1↑2↓	0↑1↓	+0.7	+0.4
Bile acid metabolism	+0.1	+1.0	+1.4	+2.6*	−0.9	+0.1	−1.5	−0.5	0.351	1↑1↓	1↑0↓	−2.5	+2.2
Protein secretion	+0.4	−2.5*	+0.8	−0.6	+0.7	−0.5	−0.8	−1.2	0.351	0↑0↓	0↑0↓	+2.7	−2.8

≤ −8 −4 0 +4 ≥ +8
set z (color clipped at ±8)

Cells: z of the set against genes permuted within identity and expression bins of the tissue; * $q < 0.05$; gray cell, fewer than 15 genes. Right: q, recurrence across tissues; S_g and profile correlation, tissues with the set above (↑) and below (↓); π and dN/dS, mean z over tissues. Gray type: not recurrent on Δα (q), on that readout (S_g, profile correlation), or |mean z| below 1.96 (π, dN/dS).